

Homework 5

- 1 Einstein gravity with a cosmological constant has the action

$$S = \frac{1}{16\pi G} \int dx^D \sqrt{-g} (R - 2\Lambda) \quad (1)$$

The equation of motion is

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -\Lambda g_{\mu\nu} \quad (2)$$

where the right hand side can be viewed as a stress tensor. For both negative and positive cosmological constants, please check if the weak, strong, null, dominant energy conditions are satisfied.

- 2 Consider the BTZ black hole

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2(d\phi + \frac{r_+r_-}{lr^2}dt)^2, \quad \phi \sim \phi + 2\pi \quad (3)$$

where

$$f(r) = \frac{(r^2 - r_-^2)(r^2 - r_+^2)}{l^2 r^2}. \quad (4)$$

- i Check that BTZ black holes are solutions to the Einstein gravity with cosmological constant, and write down the relation between Λ and parameters appearing in the solution
- ii Find the horizon generator, and work out the Hawking temperature.
- iii Draw the Penrose diagram for the BTZ black hole.

- 3 Find the event horizon and apparent horizon for Vaidya-AdS in 3d,

$$\frac{ds^2}{l^2} = -(r^2 - m(v))dv^2 + 2dvdr + r^2 dx^2 \quad (5)$$

- 4 (Witten's notes 1901.03928 P54-56) Consider de Sitter space M

$$d\vec{s}^2 = -dt^2 + R^2 e^{-\frac{2t}{R}} d\vec{x}^2, \quad \vec{x} \in R^{D-1} \quad (6)$$

and also the global de Sitter $\hat{M}$,

$$ds^2 = -dt^2 + R^2 \cosh^2(t/R) d\Omega^2 \quad (7)$$

- i (M, g) is globally hyperbolic with **noncompact** Cauchy surface $S : t = 0$. Find the compact trapped surface.
 - ii Penrose theorem applies in this case with (M, g) . Find the incomplete null geodesic. Tell what happens. Is there a singularity or the geodesic escaping from M ?
 - iii* Show that global dS $\hat{M}$ is globally hyperbolic with **compact** Cauchy surface $\hat{S}$, and convince yourself that $(\hat{M}, \hat{g})$ is geodesic complete.
- *5 (Optional question; need result of week 1)
- A is an interval in the boundary of AdS_{d+1} with end points $P_{\pm} = (t = 0, x = \pm \frac{L}{2})$. $D(A)$ is the domain of dependence at the boundary. γ is the Ryu-Takayanagi surface, which is the spacelike geodesic with end points P_+ and P_- (question 1 in homework 1).
- (a) The entanglement wedge $W(A)$ is the domain of dependence of any achronal bulk surface whose boundary is $\gamma \cup A$.
 - (b) The causal wedge $C(A)$: $J^+(D(A)) \cap J^-(D(A))$ in the bulk.
- Check that $W(A) = C(A)$ for AdS_3 .